

KALTON–ROBERTS STABILITY ANSWERS AN INTERNAL PROBABILITY LIFTING QUESTION

AGENT LANE 15

ABSTRACT. Duanmu, Schritterser, and Weiss ask whether internal almost probability measures are close to internal probability measures, and whether the internal almost probability appearing in their general Loeb-extension theorem can be replaced by a genuine internal probability. The answer is yes, using the Kalton–Roberts stability theorem for nearly additive set functions. This settles these auxiliary lifting questions, though not the separate problem of proving $\mathcal{H} \subset \overline{\mathcal{F}}$ for arbitrary Loeb-equivalent internal probability spaces.

1. SOURCE QUESTIONS

The source paper is H. Duanmu, D. Schritterser, and W. Weiss, *Loeb Extension and Loeb Equivalence II*, arXiv:2112.13955. After proving an internal-almost-probability version of the general extension theorem, the paper asks whether every internal almost probability space is close to an internal probability space, and then asks whether the almost probability in Theorem 3.5 can be replaced by an internal probability measure.

In this section, we give a partial answer to Question 1 when $\mathcal{M}$ and $\mathcal{N}$ are general internal probability spaces. We start with the following definition:

Definition 2.1. Let X be an internal space equipped with an internal algebra $\mathcal{A}$. An internal almost probability measure P is an internal mapping from $\mathcal{A}$ to ${}^*[0, 1]$ such that:

- (1) $P(\emptyset) = 0$ and $P(X) = 1$;
- (2) For every $A, B \in \mathcal{A}$, $P(A \cup B) \approx P(A) + P(B)$.

Note that internal almost probability spaces may not be internal probability spaces. One natural question to ask is:

Question 5. *Is every internal almost probability space “close” to some internal probability space?*

As is well-known, Loeb probability measures are constructed from internal probability measures. The next lemma shows that we can construct Loeb probability measures from internal almost probability measures. The proof is almost identical to the proof of the existence of the Loeb measure (see [ACH97, Section. 4, Theorem. 2.1]). We give a full proof for completeness.

Lemma 2.2. *Let $(X, \mathcal{A}, P)$ be an internal almost probability space. Then there exists a standard countably additive probability space $(X, \overline{\mathcal{A}}, \overline{P})$ such that:*

- (1) $\overline{\mathcal{A}}$ is a σ -algebra with $\mathcal{A} \subset \overline{\mathcal{A}} \subset \mathcal{P}(X)$;
- (2) $\overline{P} = \text{st}(P)$ on $\mathcal{A}$;
- (3) For every $A \in \overline{\mathcal{A}}$ and every standard $\epsilon > 0$, there exist $A_i, A_o \in \mathcal{A}$ such that $A_i \subset A \subset A_o$ and $P(A_o \setminus A_i) < \epsilon$;
- (4) For every $A \in \overline{\mathcal{A}}$, there is a $B \in \mathcal{A}$ such that $\overline{P}(A \triangle B) = 0$

Proof. Note that $(X, \mathcal{A}, \text{st}(P))$ is a finitely additive probability space. Let $A_0 \supset A_1 \supset A_2 \dots \supset A_n \dots$ be a countable non-increasing sequence of elements in $\mathcal{A}$ such that $\bigcap_{n \in \mathbb{N}} A_n = \emptyset$. By saturation, $A_N = \emptyset$ for all $N \in {}^*\mathbb{N} \setminus \mathbb{N}$. Thus, we have $\lim_{n \rightarrow \infty} \text{st}(P)(A_n) = \text{st}(P)(A_N) = 0$. So the first three items follow from the

The displayed definition of internal almost probability in the source omits the word “disjoint” in its finite-additivity condition. The literal all-pairs reading is inconsistent, since $A = B = X$ would give $1 \approx 2$. Throughout this note we use the intended disjoint version:

$$A \cap B = \emptyset \implies P(A \cup B) \approx P(A) + P(B).$$

2. KALTON–ROBERTS INPUT

Theorem 1 (Kalton–Roberts). *There is a universal constant K with the following property. Let $\mathcal{A}$ be an algebra of subsets of a set X , and let $f : \mathcal{A} \rightarrow \mathbb{R}$ satisfy*

$$|f(A \cup B) - f(A) - f(B)| \leq \varepsilon$$

whenever $A, B \in \mathcal{A}$ are disjoint. Then there is a finitely additive signed measure $m : \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\sup_{A \in \mathcal{A}} |f(A) - m(A)| \leq K\varepsilon.$$

This is Theorem 4.1 of Kalton and Roberts [2]. The following positive version is the form needed for Loeb spaces.

Corollary 1 (Positive normalized form). *There is a universal constant C such that, whenever $f : \mathcal{A} \rightarrow [0, 1]$ satisfies $f(\emptyset) = 0$, $f(X) = 1$, and is ε -additive on disjoint pairs, there is a finitely additive probability measure $p : \mathcal{A} \rightarrow [0, 1]$ such that*

$$\sup_{A \in \mathcal{A}} |f(A) - p(A)| \leq C\varepsilon.$$

Proof. Apply Theorem 1 to obtain a signed charge m with $\|m - f\|_\infty \leq K\varepsilon$. Then $m(A) \geq -K\varepsilon$ for every A , and $|m(X) - 1| \leq K\varepsilon$. By the Jordan decomposition of a bounded finitely additive signed measure, $m = m^+ - m^-$, and

$$m^-(X) = \sup_{A \in \mathcal{A}} (-m(A)) \leq K\varepsilon.$$

Hence $s := m^+(X) = m(X) + m^-(X)$ satisfies $s = 1 + O(\varepsilon)$, and for small enough ε the normalized charge $p = m^+/s$ is a finitely additive probability. Moreover

$$|p(A) - m(A)| \leq |s - 1| + m^-(X) \leq C_1\varepsilon$$

for a universal C_1 , uniformly in A . Combining this with $\|m - f\|_\infty \leq K\varepsilon$ gives the result. The remaining large- ε case is absorbed by increasing C . $\square$

3. NONSTANDARD CONSEQUENCE

Theorem 2. *Let $(X, \mathcal{A}, P)$ be an internal almost probability space, with approximate additivity understood on disjoint pairs. Then there is an internal finitely additive probability measure $Q : \mathcal{A} \rightarrow {}^*[0, 1]$ such that*

$$Q(A) \approx P(A) \quad (A \in \mathcal{A}).$$

Proof. The assertion that P is almost additive says that for every standard $\varepsilon > 0$,

$$|P(A \cup B) - P(A) - P(B)| \leq \varepsilon$$

for all disjoint $A, B \in \mathcal{A}$. By underspill, there is a positive infinitesimal δ such that this inequality holds with δ in place of every standard ε . Transfer Corollary 1 to the internal algebra $\mathcal{A}$ and the internal set function P . We obtain an internal finitely additive probability measure Q on $\mathcal{A}$ satisfying

$$\sup_{A \in \mathcal{A}} |Q(A) - P(A)| \leq C\delta.$$

Since $C\delta$ is infinitesimal, $Q(A) \approx P(A)$ for every $A \in \mathcal{A}$. $\square$

Corollary 2. *Let $(\Omega, \mathcal{F}, \mu)$ be an internal probability space, let $\mathcal{G}$ be an internal algebra on Ω , and let $\mathcal{H}$ be the internal algebra generated by $\mathcal{F} \cup \mathcal{G}$. If*

$$\mathcal{H} \subset \overline{\mathcal{F}},$$

then there is an internal probability measure Q on $\mathcal{H}$ such that

$$(\Omega, \mathcal{H}, Q)$$

is Loeb equivalent to $(\Omega, \mathcal{F}, \mu)$.

Proof. Theorem 3.5 of the source paper constructs an internal almost probability measure P on $\mathcal{H}$ whose Loeb extension is equivalent to $(\Omega, \mathcal{F}, \mu)$. Apply Theorem 2 to obtain an internal probability Q on $\mathcal{H}$ with $Q(H) \approx P(H)$ for all $H \in \mathcal{H}$. The Loeb constructions from P and Q therefore agree on $\mathcal{H}$, and hence have the same completed Loeb measure space. Thus $(\Omega, \mathcal{H}, Q)$ is Loeb equivalent to $(\Omega, \mathcal{F}, \mu)$. $\square$

Remark 1. *Corollary 2 answers Question **internalapprox** from the source paper. It does not settle the full arbitrary-algebra version of Keisler–Sun Question 3, because that still requires proving the hypothesis $\mathcal{H} \subset \overline{\mathcal{F}}$ from Loeb equivalence of $(\Omega, \mathcal{F}, \mu)$ and $(\Omega, \mathcal{G}, \nu)$.*

4. NOVELTY CHECK

Local run indexes were searched for arXiv:2112.13955, Loeb equivalence, internal almost probability, Kalton–Roberts, and nearly additive set functions. No prior packet was found. A bounded web search on 2026-07-19 found the source paper and references to the Kalton–Roberts theorem, but no explicit later Loeb-theoretic answer to the two auxiliary questions treated here.

REFERENCES

- [1] H. Duanmu, D. Schritterser, and W. Weiss, *Loeb Extension and Loeb Equivalence II*, arXiv:2112.13955.
- [2] N. J. Kalton and J. W. Roberts, *Uniformly exhaustive submeasures and nearly additive set functions*, Trans. Amer. Math. Soc. 278 (1983), no. 2, 803–816. doi:10.1090/S0002-9947-1983-0701524-4.